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INSULATION CO-ORDINATION

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TRANSMISSION ENGINEERING STANDARD

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1.0 PURPOSE AND SCOPE

This Standard indicates the guidelines which have been followed for selection of insulation levels for insulation coordination of substation equipment in shielded substation in SEC System.

SEC substations shall be designed for the equipment insulation levels specified in 01-TMSS-01.

The purpose of insulation co-ordination shall be to co-relate the insulation withstand levels of protected equipment and circuits with the protection characteristics of surge arresters, such that the insulation is protected from over voltages with overall economy. Clearances are to be determined in line with TES-P-119.08 by applying the principles and practices of insulation co-ordination based on following three main elements:

- The knowledge of the voltage stresses which may occur at the work-site.
- The knowledge of the electrical strength of the work site insulation when submitted to such voltage stresses.
- The assessment of the probability of occurrence of insulation failures in the considered situation of voltage stresses and electrical strength.

Insulation coordination shall be verified correlating the internal dielectric strength of electrical equipment and the characteristics and location of protective devices with expected types of overvoltages. Electrical strength reduction of the external insulation due to worst case atmospheric humidity/pollution and changes in dielectric strengths due to changes in altitude with corresponding change of air density shall be considered wherever applicable as specified in 01-TMSS-01. To avoid derating, suitable higher withstand level shall be chosen per 01-TMSS-01. In case of combined voltage test, the atmospheric correction factor should be applied to the total test voltage, which is the sum of the two components.

The reduction of electrical strength due to various perturbing factors in relation to live line maintenance operations are not covered in this chapter.

If rod gaps are used at 230kV and 132kV or 115kV across the line-entrance insulator stack, where impulse withstand voltage of the line-to-ground insulation gets increased significantly as a result of insulator leakage requirement, the lightning and switching impulse tests shall be performed on the actual gap configuration and geometry to be used, at minimum withstand levels specified in 01-TMSS-01. Surge arresters shall be installed at overhead line entrance to substation and underground cable termination, high voltage SF6-to-air termination, both ends of long length of underground cable rated 110kV and above, capacitor bank connected with series reactance and with the shunt reactor with proper insulation co-ordination.

The actual testing/test reports of the equipment to determine their performance for insulation co-ordination shall be given maximum importance. It is to be ensured from the test reports

that the disconnect switch open gap has a higher insulation withstand level than the insulation to ground with the creepage distance specified in 01-TMSS-01.

Insulation co-ordination study of the GIS shall be performed separately to ensure the adequacy of protective margin (considering very fast transients also), location and number of surge arresters to be provided in the GIS.

2.0 SYSTEM OVERVOLTAGES

Following overvoltages stressing dielectric strength of insulation shall be considered:

- Power frequency overvoltages (permanent and temporary low frequency)
- Switching overvoltages (slow front impulse, positive polarity at dry condition)
- Lightning overvoltages (fast front impulse)

2.1 Temporary Power Frequency Overvoltages

Temporary overvoltages on healthy phases due to faults to ground, circuit backfeeding, sudden load rejection (when the load is disconnected at the end of a long transmission line, or when switching-off a large inductive load), resonance or ferroresonance [when circuits with capacitive elements (lines, cables, shunt capacitors) and inductive elements (unloaded transformers, shunt reactors) having nonlinear magnetizing characteristics are energized with the long line/cable] and other system contingencies shall be evaluated. For an effectively grounded system, the coefficient of grounding shall be taken as 80%. The worst overvoltages due to system resonance, looking from transformer location, at second, third, fourth or fifth harmonic shall be evaluated wherever necessary. For resistance/reactor grounded system, the coefficient of grounding shall be separately evaluated to determine the requirements of insulation co-ordination.

2.2 Switching Overvoltages

Switching overvoltage caused by long line switching, high speed auto-reclosing, out of phase switching of cable circuits/capacitor banks/shunt reactors, circuit breaker restriking, load rejection, current chopping etc. shall be evaluated for 230kV and 380kV. The standard switching impulse considered against switching overvoltages shall be a full 250/2500µs impulse having a front time of 250µs and a tail time (time-to-half value) of 2500µs with the peak value equivalent to Basic Switching Impulse Insulation Level (BSL) mentioned in 01-TMSS-01.

Switching overvoltages resulting from line energisation with trapped charges and high speed reclosing and voltage stresses caused by very fast transients (rise time of 3-10ns) due to switching (worst case) in Gas Insulated Substation (GIS) shall be evaluated separately wherever applicable.

2.3 Lightning Overvoltages

Lightning overvoltages caused by direct strokes to phase conductors or induced lightning surges due to back flashovers and strokes to the earth very close to a line

are to be evaluated with respect to system impedance and lightning stroke current. In shielded substations, the value of lightning discharge currents shall be considered as 10kA for SEC-EOA and SEC-COA area and 20kA for SEC-WOA and SEC-SOA area. The standard lightning impulse considered against lightning overvoltages shall be a full 1.2/50µs impulse having a front time of 1.2 µs and a tail time (time-to-half value) of 50 µs with the peak value equivalent to Basic Lightning Impulse Insulation Level (BIL) mentioned in 01-TMSS-01.

3.0 PROTECTION LEVEL

Bushings of dead tank circuit breaker, instrument transformers and substation supporting insulators in a bay shall be protected from excessive overvoltages by the bay surge arrester connected on the power transformer. Details of the maximum protective level of surge arrester (in terms of residual voltage) corresponding to certain discharge current and basis of their selection shall be per TES-P-119.06. Insulation level for capacitor bank support insulators, etc.shall be so selected that sufficient protective margin exists between the maximum overvoltage and the minimum dielectric strength. For voltage levels up to 132kV switching impulse is not decisive for insulation breakdown. Hence, switching impulse withstand levels shall not be considered for voltage levels up to 132kV. Areas, where Isokeraunic level is more than 10 stormdays/year, use of additional surge arresters and their locations shall be carefully evaluated.

4.0 PROTECTIVE RATIOS

Unless otherwise specified, the minimum Protective Ratio between required insulation level (BIL or BSL) and maximum protective level of protective devices shall be as specified below:

Minimum Protective Ratio for lightning surge withstand level is

$$= \frac{\text{Required BIL}}{\text{Maximum lightning impulse protection level of protective device}} \geq 1.20$$

Minimum Protective Ratio for switching surge withstand level for 230kV and 380kV system is

$$= \frac{\text{Required BSL}}{\text{Maximum switching impulse protection level of protective device}} \geq 1.15$$

The maximum lightning impulse protection level of protective device shall include $5kV_{peak}$ drop of 3m lead between ground terminal of arrester and the insulated bushing of the surge counter. The minimum protective ratio for lightning impulse withstand level shall be taken as 1.4 for voltage levels upto 230kV and 1.2 for 380kV systems.

5.0 SELECTION OF INSULATION LEVELS

5.1 Selection of Equipment Power Frequency Withstand Voltages

Once the maximum power frequency overvoltages is decided, the next higher standard power frequency withstand voltage shall be selected from respective IEC/ANSI/IEEE standards for internal insulation of equipment such as transformers, circuit breakers, etc. adopting the most conservative approach unless otherwise the same is specified in 01-TMSS-01.

For external insulation, the wet (10s) power frequency withstand voltages as specified in IEC/ANSI/IEEE Standards for outdoor bushings shall be selected unless otherwise the same is specified in 01-TMSS-01. Power frequency excitation voltage of a contaminated external insulation shall dictate its creepage distance per 01-TMSS-01.

For resistance/reactor grounded (non-effectively grounded) existing system, the internal and external equipment power frequency withstand voltage shall be separately evaluated to determine the rated voltage of the surge arrester.

5.2 Selection of Equipment BIL

For equipment internal BIL, value of the maximum lightning impulse protection level of protective device shall be multiplied by the protective ratio. Then, the next higher available BIL from respective IEC/ANSI/IEEE standards for equipment such as transformers, circuit breakers, disconnect switch, etc. shall be selected adopting the most conservative approach unless otherwise the same is specified in 01-TMSS-01.

If surge arresters are used in the OLTC of the power transformer, the BIL testing of the power transformer shall be dictated by relevant IEC/ANSI/IEEE standards.

External BIL shall be adopted as one step higher than the selected internal BIL taking into consideration the effects of atmospheric and other external conditions such as pollution and humidity. Specified creepage distance shall be efficiently utilized to obtain the required BIL/increased dry flashover voltage.

For resistance/reactor grounded (non-effectively grounded) existing system, the maximum protective level of surge arrester (in terms of residual voltage) for lightning impulse shall be considered per TES-P-119.06 to determine the equipment internal BIL.

5.3 Selection of Equipment BSL for 380kV Systems

The value of the maximum switching impulse protection level of protective device shall be multiplied by the protective ratio of 1.15. Then, the next higher standard BSL from respective IEC/ANSI/IEEE standards shall be adopted unless otherwise the same is specified in 01-TMSS-01.

5.4 Selection of Equipment Chopped Impulse Withstand

When requested, (in case of SF_6-to-Oil transformer termination) chopped (at 3µs) lightning impulse (1.2/50µs, 1.1 times BIL) tests shall be performed on outdoor power transformer per IEC 60076-3. When requested, chopped lightning impulse tests shall be performed on all outdoor current transformers upto 230kV per IEC 60044-1. When requested, multiple chopped lightning impulse tests shall be performed on all outdoor oil-filled 380kV current transformers per IEC 60044-1. When requested, chopped (at 2µs) lightning impulse (1.2/50µs, 1.29 times BIL) tests and chopped (at 3µs) lightning impulse (1.2/50µs, 1.15 times BIL) tests shall be performed on high voltage circuit breaker for reactor switching per IEC 61233.

The following example may be followed to apply the altitude (atmospheric) correction factors on the BIL and Power Frequency Withstand Voltage to ground for the air-insulated equipment per formula given in 01-TMSS-01, Rev.0:

Example-1:
The required BIL and Power Frequency Withstand Voltage to ground for 230kV outdoor station post insulator have been specified as 1050kV and 460kV respectively in 01-TMSS-01, Rev.0 for an altitude within 1000m of mean sea level. Determine the required BIL and Power Frequency Withstand Voltage to ground for the same outdoor station post insulator at an altitude of 2000m.

The altitude correction factor, $K_a = e^{m\left(\frac{H}{8150}\right)}$

Where,

m = 1.0 for co-ordination lightning impulse withstand voltage and

m = 0.5 for short-duration power frequency withstand voltages for normal insulators and

H = 2000m

Applying above values, K_a to be multiplied with BIL = $e^{1.0\left(\frac{2000}{8150}\right)}$ = 1.278 and

the required BIL at an altitude of 2000m shall be 1341.9 kV.

Also, applying above values, K_a to be multiplied with Power Frequency Withstand Voltage = $e^{0.5\left(\frac{2000}{8150}\right)}$ = 1.130 and the required PFWV at an altitude of 2000m shall be 519.8 kV.

Hence, next higher Standard BIL of 1425kV and Power Frequency Withstand Voltage of 570kV shall be provided for the same outdoor station post insulator at an altitude of 2000m.

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